

Cooperative games

1. A characteristic function game (N, v) assigns ...
 - (a) to each coalition of players a worth v
 - (b) to each player a single action
 - (c) a probability to each outcome
 - (d) to each player a payoff matrix
2. The Shapley value of a player is ...
 - (a) their average marginal contribution over all orderings of the players
 - (b) their payoff in the worst coalition
 - (c) the worth of the grand coalition divided equally
 - (d) the largest coalition they belong to
3. The efficiency property of the Shapley value says that ...
 - (a) the players' payoffs sum to the worth of the grand coalition
 - (b) every player receives the same amount
 - (c) the strongest player gets everything
 - (d) payoffs are always non-negative
4. A null player is one who ...
 - (a) adds nothing to any coalition they join, and so receives a Shapley value of 0
 - (b) has the highest marginal contribution
 - (c) always receives the most
 - (d) is in no coalition
5. The symmetry property says that two players who ...
 - (a) join last always tie
 - (b) have the same name receive equal payoffs
 - (c) are both null players receive everything
 - (d) contribute equally to every coalition receive equal Shapley values
6. The marginal contribution of player i to a coalition S is ...
 - (a) $v(S)/|S|$
 - (b) $v(\{i\})$
 - (c) $v(S \cup \{i\}) - v(S)$
 - (d) $v(S \cup \{i\})$

7. The grand coalition is ...
 - (a) the coalition of all the players
 - (b) the empty coalition
 - (c) the strongest single player
 - (d) any coalition of size two
8. A payoff vector must satisfy ...
 - (a) $\sum_i \lambda_i = 0$
 - (b) $\lambda_i = 1$ for all i
 - (c) $\lambda_i = v(\{i\})$
 - (d) $\sum_i \lambda_i = v(N)$, sharing out the grand coalition's worth
9. The additivity property of the Shapley value says that ...
 - (a) marginal contributions are ignored
 - (b) all players receive equal payoffs
 - (c) the grand coalition gets nothing
 - (d) the value of a sum of two games is the sum of the separate Shapley values
10. The Shapley value is given by the formula ...
 - (a) $\phi_i = \frac{1}{N!} \sum_{\pi} \Delta_{\pi}(i)$
 - (b) $\phi_i = v(N)/N$
 - (c) $\phi_i = v(\{i\})$
 - (d) $\phi_i = \max_S v(S)$
11. By convention, the worth of the empty coalition is ...
 - (a) undefined
 - (b) $v(\emptyset) = 0$
 - (c) $v(\emptyset) = 1$
 - (d) $v(\emptyset) = v(N)$
12. The number of orderings of N players, over which the Shapley value averages, is ...
 - (a) N
 - (b) N^2
 - (c) 2^N
 - (d) $N!$
13. Two players who contribute equally to every coalition ...

- (a) split the whole worth between them
 - (b) receive nothing
 - (c) are both null players
 - (d) receive equal Shapley values (symmetry)
14. The Shapley value is the unique allocation satisfying ...
- (a) no particular axioms
 - (b) efficiency alone
 - (c) symmetry alone
 - (d) efficiency, the null player property, symmetry and additivity
15. The characteristic function $v(S)$ gives ...
- (a) the Shapley value of S
 - (b) the worth that coalition S can secure on its own
 - (c) the payoff to a single player
 - (d) the number of players in S
16. The set of predecessors $S_\pi(i)$ of player i in an ordering π is ...
- (a) the grand coalition
 - (b) the players who appear before i in π
 - (c) all players other than i
 - (d) the players who appear after i in π
17. A characteristic function game is superadditive if, for disjoint coalitions S and T , ...
- (a) $v(S \cup T) \geq v(S) + v(T)$
 - (b) $v(S \cup T) \leq v(S) + v(T)$
 - (c) $v(S) = v(T)$
 - (d) $v(S \cup T) = 0$
18. A characteristic function game is monotone if ...
- (a) $v(S) = |S|$ for all S
 - (b) $v(S) \geq v(T)$ whenever $S \subseteq T$
 - (c) $v(\emptyset) = 1$
 - (d) $v(S) \leq v(T)$ whenever $S \subseteq T$
19. Written as a sum over coalitions, the Shapley value weights the marginal contribution of i over S by ...
- (a) $\frac{|S|}{N}$

(b) $\frac{1}{N}$

(c) $\frac{|S|! (N-|S|-1)!}{N!}$

(d) $\frac{1}{N!}$

20. The Shapley value is the unique payoff vector satisfying efficiency, symmetry, additivity and ...

(a) superadditivity

(b) the null player property

(c) monotonicity

(d) individual rationality